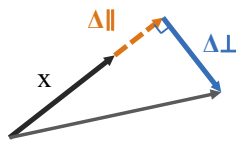


Geometric Decomposition



$$\Delta = \Delta_{\parallel} + \Delta_{\perp}$$

$$\Delta_{\parallel} = \alpha x$$

$$\Delta_{\perp} \perp x$$

Scaling-like

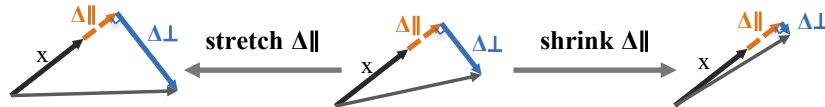
$\Delta_{\parallel}$ change magnitude

Direction-changing

$\Delta_{\perp}$ change direction

Manipulate the perpendicular component

Intervention: stretch or shrink $\Delta_{\perp}$ while keeping the rest of the update fixed



Changing $\Delta_{\perp}$ sharply hurts performance.

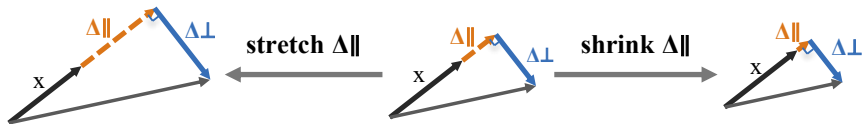


Performance



Manipulate the parallel component

Intervention: scale $\Delta_{\parallel}$ up or down while keeping the rest of the update fixed



Changing $\Delta_{\parallel}$ has limited impact on performance.

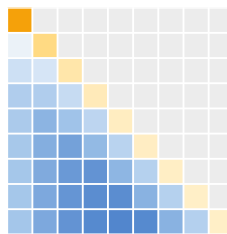


Performance

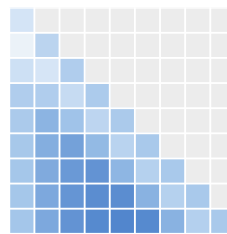


Attention / Value-Based Interpretation

Baseline attention



Diagonal-suppressed



Deactivate



Diagonal suppression weakens self-copy and redistributes mass off diagonal